



*Collaboration between Dan School of Drama and
Music and Queen's Computer Science*

2025/01/06 to 2025/04/09

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INTRODUCTION

Caryl Churchill's *Love and Information* (2012) presents a unique theatrical challenge. The play consists of seven sections containing over 60 short scenes, which are intended to be performed in different arrangements each night.

For Queen's University's production, an algorithm was developed to generate a different scene order for each performance. The goal was to respect structural constraints in the script while integrating meaningful randomness and audience influence. Specifically, the challenge was to incorporate daily questionnaire responses into the structure of each night's show, making every performance uniquely shaped by the audience.

WEBSITE DESIGN

WEBSITE ARCHITECTURE

The production website consisted of four core pages:

1. **Homepage** (Figure 1 - Love and Information Home Page)
A visually responsive landing page introducing the project. This page featured embedded SVG assets and auto-scaling text to match various devices and screen sizes. We also included a "Scroll to Ticket" button for easy navigation.
2. **Questionnaire** (Figure 2 – Questionnaire Design)
Audience members were invited to participate in an interactive 10-question survey prior to ticket purchase. The quiz was styled with custom HTML/CSS and powered by JavaScript, which collected answer data and asynchronously submitted it to the server via AJAX to populate the MySQL `quiz_responses` table.
3. **Cast & Crew Dashboard** (Figure 3 - Scene Order List)
An internal-access-only page was created where crew members could download the day's generated scene order. This list was refreshed 4 hours before each performance, using a PHP backend script triggered by a daily automation.
4. **Trait Visualization Page** (Figure 4 - Questionnaire Visualization)
An aesthetic front-facing visual dashboard was also created, displaying a **pie chart** of the current audience's personality trait breakdown. This helped contextualize the performance for both audience and cast, reinforcing the interactivity of the experience.

DESIGNS

HOME PAGE



Figure 1 - Love and Information Home Page

AUDIENCE QUESTIONNAIRE - DESIGN

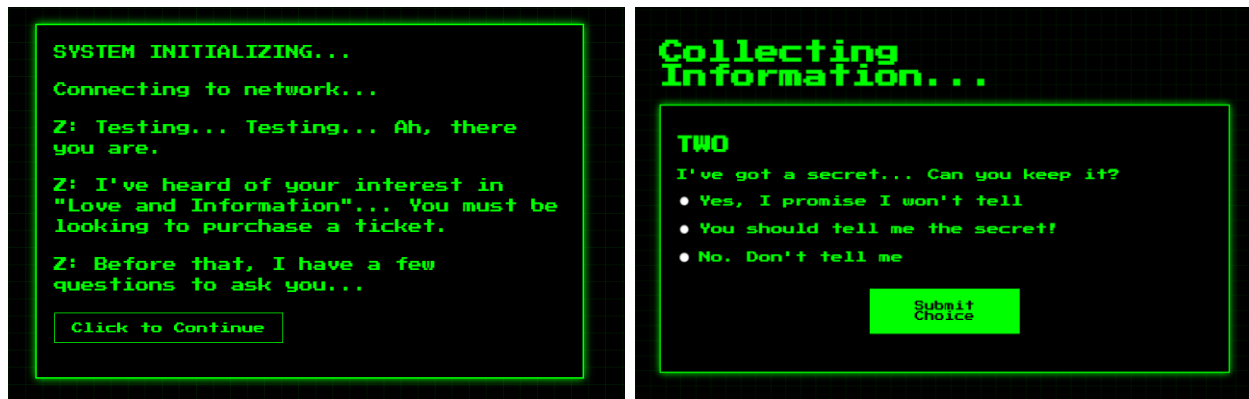


Figure 2 – Questionnaire Design

CAST AND CREW DASHBOARD

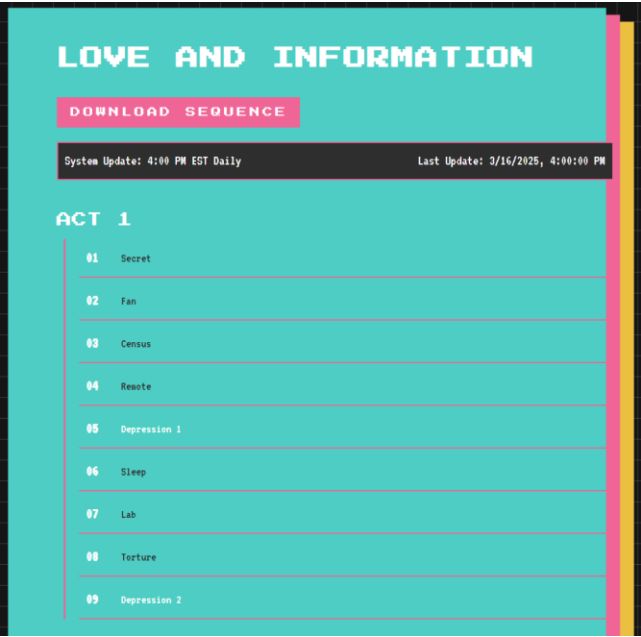


Figure 3 - Scene Order List

VISUALIZATION PAGE

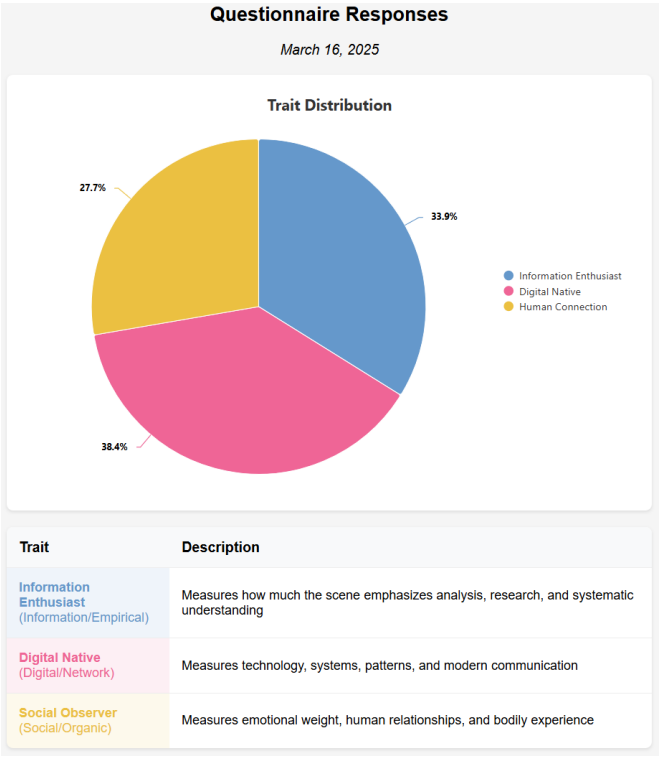


Figure 4 - Questionnaire Visualization

DESIGN AND USER EXPERIENCE

The website for *Love and Information* was intentionally crafted not just as a tool, but as an extension of the play's aesthetic and philosophical themes — bridging technology, memory, and human connection. Each section of the site was given a unique visual identity that aligned with either the artistic tone of the show or the branding vision set by the production's marketing team.

HOMEPAGE – BRANDED & IMMERSIVE

The homepage featured **SVG graphics directly adapted from the marketing team's official poster art**, ensuring immediate brand continuity and visual coherence across print and web.

Responsive layout techniques were used so that text and imagery would scale gracefully on mobile and desktop. The landing design aimed to reflect the modular, fragmented tone of the play, using sharp geometric shapes, subtle animations, and soft fades to evoke the sensation of information flow.

PERSONALITY QUESTIONNAIRE

To inject playfulness and intrigue into the audience experience, the questionnaire was **styled after 1980s arcade pixel games**. The UI resembled a vintage text-based adventure game, with:

- A **black and neon green** theme reminiscent of early Matrix-style command prompts
- Typewriter-effect animations and blinking cursors

DASHBOARD AND VISUALIZATION PAGE

The cast & crew dashboard and audience visualization page adopted a **triple-tone design strategy**:

- Colour schemes pulled directly from the **production's marketing color guide**, ensuring consistency with design guidelines.
- The dashboard's design took inspiration from **early 90s pop cartoon aesthetics**, featuring:
 - **Triple-layered background gradients** for a 3D feel
 - Bold pastel section blocks and Asymmetrical layout with Pixelated font

This playful, comic-book-inspired palette provided a sharp contrast to the minimalist questionnaire page to indicate a shift in eras similar to the technological themes portrayed in the play.

VISUAL HIERARCHY AND ACCESSIBILITY

The entire site used:

- **Large headers and high-contrast colors** to ensure readability from afar
- Fully responsive layouts for cast viewing on phones backstage
- Color-coded legends and tooltips to help decode pie charts and data outputs

ALGORITHM

The system is centered around audience interactivity: a 10-question personality quiz taken by audience members before each show. Responses to this survey were processed to determine a daily audience profile, which in turn influenced the ordering of scenes. The full pipeline runs automatically and publishes an updated ordering each day before showtime.

REQUIREMENTS

The algorithm needed to satisfy several key constraints:

1. Daily Variation: A new scene ordering each day, updated ~4 hours before the show.
2. Structural Requirements: The play's scenes needed to follow specific organizational rules:
 - Seven fixed acts.
 - Certain scenes fixed to specific acts (e.g., "Census" must appear in Act 1).
 - Some scenes fixed in position (e.g., "Facts" as the final scene).
 - Ensure distribution of "Depression" scenes across predetermined act locations.
3. Audience Influence: Audience input from a pre-show questionnaire must guide scene selection and ordering.
4. Trait Categorization: Each response maps to a set of three personality traits:
 - Digital Native (DN): Comfort with modern digital systems.
 - Social Observer (SO): Empathy, social relationships, emotional connection.
 - Information Enthusiast (IE): Analytical, research-driven thinking.

RESPONSE COLLECTION AND AGGREGATION

Each audience member completed a 10-question personality quiz. Each of the 10 survey questions had three possible answers (as shown in Figure 5 - Questionnaire) and contributed points to the traits DN, IE, and SO. For example:

- Q2: "I have a secret. Can you keep it?"
 - "Tell me the secret!" → +1 IE
 - "You can trust me" → +1 DN
 - "Don't tell me" → +1 SO

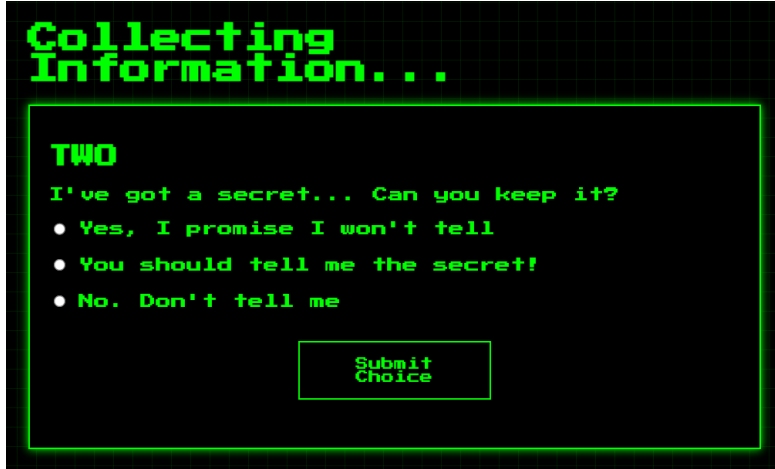


Figure 5 - Questionnaire

The answer choices were mapped as a triple:

$$A_i = (dn_i, ie_i, so_i), \text{ where } i \in [1, N]$$

with $N = (\text{Number of people who took the quiz}) \times 10$ as each respondent answered 10 questions.

To generate an audience profile, we computed the average score across all responses:

$$DN_\mu = \frac{1}{N} \sum_{i=1}^N DN_i, \quad IE_\mu = \frac{1}{N} \sum_{i=1}^N IE_i, \quad SO_\mu = \frac{1}{N} \sum_{i=1}^N SO_i$$

Which forms the **audience triple**:

$$A = (DN_\mu, IE_\mu, SO_\mu) = \left(\frac{1}{N} \sum_{i=1}^N DN_i, \frac{1}{N} \sum_{i=1}^N IE_i, \frac{1}{N} \sum_{i=1}^N SO_i \right)$$

INITIAL SCENE-PERSONALITY MAPPING AND ANALYSIS

To create meaningful connections between audience input and the scene ordering of Love and Information, we began by mapping each of the 60+ scenes to thematic clusters aligned with our personality framework. Each scene was annotated with up to three tags, corresponding to the following audience personality types.

Each scene was rated on a scale from **0 to 5** in each category, based on the following scoring guide:

Score	IE (Information)	DN (Digital)	SO (Social)
5	<i>Strong emphasis on research, academic context, or systematic analysis</i>	<i>Central role of digital systems, modern platforms, or technical networks</i>	<i>Deep emotional impact, relationships, and bodily experience at the core</i>
3-4	<i>Information present but secondary</i>	<i>Tech/digital elements support the main theme</i>	<i>Emotion or relationships present but not dominant</i>
1-2	<i>Minimal analytic content; scene driven by other elements</i>	<i>No meaningful digital or technical presence</i>	<i>Primarily technical or objective, with little human/emotional weight</i>

This classification created a **scene profile vector** of the form:

$$S = (s_{DN}, s_{IE}, s_{SO}) \text{ for each scene, where } s_x \in [0,5].$$

Act	Scene	IE	DN	SO	Primary	Act	Scene	IE	DN	SO	Primary
1	Lab	5	3	2	IE	5	Linguist	5	3	2	IE
1	Census	5	3	2	IE	5	Sex	1	5	4	DN
1	Remote	3	5	2	DN	5	Rash	1	5	2	DN
1	Fan	1	5	4	DN	5	God	2	1	5	SO
1	Torture	2	1	5	SO	5	Children	2	3	5	SO
1	Sleep	2	3	5	SO	5	Shrink	3	1	5	SO
2	Terminal	3	5	2	DN	6	Climate	5	3	2	IE
2	Message	1	5	4	DN	6	Decision	2	1	5	SO
2	Grass	5	3	1	IE	6	Censor	1	4	5	SO
2	Mother	2	3	5	SO	6	Earthquake	3	5	2	DN
2	Affair	2	1	5	SO	6	TCWDKS	2	3	5	SO
2	Irrational	2	3	5	SO	6	TCWDKP	2	1	5	SO
3	God's Voice	2	1	5	SO	7	Chinese Poetry	5	3	1	IE
3	Spies	1	4	5	SO	7	Virtual	3	5	2	DN
3	Recluse	3	5	2	DN	7	Stone	2	1	5	SO
3	Schizophrenic	2	3	5	SO	7	Grief	2	3	5	SO
3	Dream	1	5	4	DN	7	Manic	1	5	4	DN
3	TCWDKF	5	3	1	IE	7	Small Thing	5	3	1	IE
4	Savant	5	3	2	IE						
4	Memory House	5	3	1	IE						
4	Wedding Video	1	5	4	DN						
4	Piano	1	5	2	DN						
4	Dinner	2	1	5	SO						
4	Flashback	3	5	1	DN						

Figure 6 - Scene Categorization

CALCULATING SCENE ORDERING

To compute how well a scene fits the audience's daily profile, we take the dot product of the audience and scene vectors:

$$W_S = A \cdot S = (DN_\mu \cdot s_{DN}) + (IE_\mu \cdot s_{IE}) + (SO_\mu \cdot s_{SO})$$

These weights W_s are then used to:

1. Prioritize scene selection per act.
2. Randomize bias so high-weight scenes appear earlier within acts.
3. Ensure variety while tailoring to audience preference.

AVERAGING RESPONSES AND ORDERING SCENES

For each performance:

- All responses for the corresponding date are retrieved from the MySQL database.
- The average trait scores are computed.
- Each scene receives a weight based on its alignment with the profile.
- Scenes are grouped by dominant trait category and shuffled internally.
- Depression scenes are inserted at specific positions.
- "Facts" is placed last.

RATIONALE AND EVOLUTION OF THE ALGORITHM

PHASE 1: NON-WEIGHTED ORDERING

The initial version of the algorithm selected scenes randomly while ensuring:

- Each act contained a fixed number of scenes.
- No scene appeared more than once.
- One or more “Depression” scenes were spaced across the acts.
- A concluding “Facts” scene always appeared at the end.

Date	Count	AVG DN	AVG IE	AVG SO	First	Second	Third
2025-03-05	36	6.17	4.83	3.28	DN	IE	SO
2025-03-06	26	5.85	5.15	3.27	DN	IE	SO
2025-03-07	12	5.67	5.08	3.08	DN	IE	SO
2025-03-08	29	5.67	5.08	3.08	DN	IE	SO
2025-03-09	14	6.43	4.14	4.07	DN	IE	SO
2025-03-10	9	6.00	4.00	4.44	DN	SO	IE
2025-03-11	7	6.43	4.43	3.57	DN	IE	SO
2025-03-12	16	6.06	4.13	4.13	DN	SO	IE
2025-03-13	8	6.25	5.88	2.50	DN	IE	SO
2025-03-14	7	6.29	3.57	5.00	DN	SO	IE
2025-03-15	13	5.54	5.62	3.38	IE	DN	SO
2025-03-16	11	6.36	4.36	3.64	DN	IE	SO
Total Averages	188.00	6.06	4.69	3.62			

Figure 7 – Non-Weighted Average Calculations

While this satisfied the requirement for a different order each day, it did not reflect audience identity in any meaningful way.

PHASE 2: WEIGHTED SCENE SELECTION

Initial survey data (Feb 19 – Mar 2, 2025) revealed imbalance of primary traits where DN dominated as the primary trait for most of the test dates when not weighted:

- **DN: 91.7%, IE: 8.3%, SO: 0%**
- **Average Scores (μ) across all days were:**
 - **IE: 4.69, DN: 6.06, SO: 3.62**

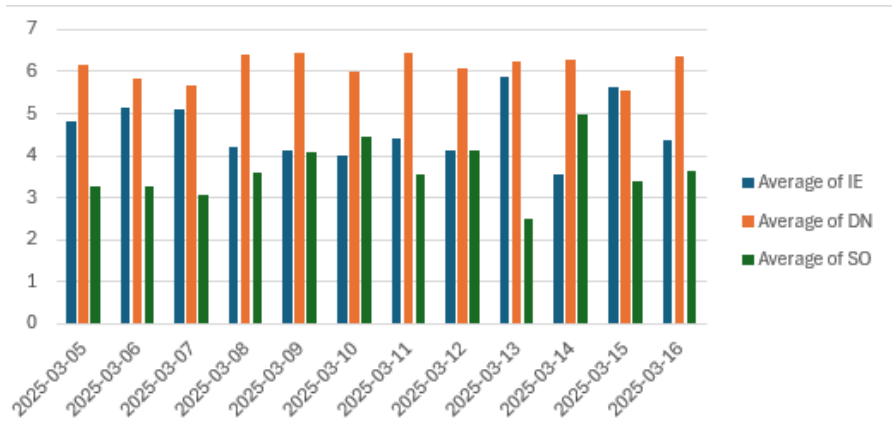


Figure 8 - Primary Traits by Dominance (Unweighted)

This heavy dominance of the Digital Native (DN) trait indicated the need for a weighting system to correct the imbalance and ensure more equitable trait representation across performances. To do this, we applied a compensatory weighting strategy to move toward an ideal of roughly 33.3% influence from each trait.

PHASE 2.1 – STATIC WEIGHTING (INITIAL MODEL)

Our first approach used a static compensatory weighting model to rebalance trait influence. We began by defining a target adjusted score **T**, representing the desired influence each trait should contribute after weighting.

Using DN as a baseline, we estimated a trial weight based on a target proportional representation of 33.3%:

$$Trial\ Weight_{DN} = \frac{0.333}{0.917} \approx 0.363$$

Applying this trial weight to DN's average score of 6.06 yielded the target adjusted score:

$$T = 6.06 \times 0.363 \approx 2.20$$

We then solved for a universal scaling constant **k**:

$$k = \frac{T \cdot \text{Trait Percentage}_{\text{DN}}}{\text{Average Score}_{\text{DN}}} = \frac{2.20 \cdot 0.917}{6.06} \approx 0.3328$$

Using this constant, we computed raw adjusted weights:

$$\text{Adjusted Weight}_{\text{DN}} = \frac{0.3328}{0.917} \approx 0.363, \quad \text{Adjusted Weight}_{\text{IE}} = \frac{0.3328}{0.083} \approx 4.01$$

Since the SO trait had 0% representation, applying the inverse formula would be undefined. Instead, a compensatory weight was assigned to match the target score directly:

$$\text{Adjusted Weight}_{\text{SO}} = \frac{T}{\text{Average Score}_{\text{SO}}} \approx 0.608$$

To make these weights more interpretable and easier to apply, we rescaled them so that the lowest weight (DN) became 1.0:

$$\text{Rescaled Weight} = \frac{\text{Adjusted Weight}}{\min(\text{All Adjusted Weights})}$$

Final rescaled weights:

$$\text{DN: } \frac{0.363}{0.363} = 1.00, \quad \text{IE: } \frac{4.01}{0.363} \approx 1.29, \quad \text{SO: } \frac{0.608}{0.363} \approx 1.67$$

In summary, we get:

Trait	Dominant Trait %	Avg Score	Raw Weight	Rescaled Weight
DN	91.7 %	6.06	0.363	1.00
IE	8.3 %	4.69	4.01	1.29
SO	0.0 %	3.62	0.608	1.67

DN Weight	IE Weight	SO Weight	Weighted AVG - DN	Weighted AVG - IE	Weighted AVG - SO	First	Second	Third
1.00	1.29	1.67	6.308	5.904	5.781	DN	IE	SO
1.00	1.29	1.67	6.077	5.954	6.038	DN	SO	IE
1.00	1.29	1.67	5.972	6.558	5.335	IE	DN	SO
1.00	1.29	1.67	6.264	5.622	5.798	DN	SO	IE
1.00	1.29	1.67	6.480	5.418	6.346	DN	SO	IE
1.00	1.29	1.67	6.200	5.418	6.903	SO	DN	IE
1.00	1.29	1.67	6.179	6.588	5.010	IE	DN	SO
1.00	1.29	1.67	6.207	5.694	6.277	SO	DN	IE
1.00	1.29	1.67	6.395	6.586	4.922	IE	DN	SO
1.00	1.29	1.67	6.581	5.618	6.303	DN	SO	IE
1.00	1.29	1.67	5.483	7.073	6.104	IE	SO	DN
1.00	1.29	1.67	5.543	6.303	6.680	SO	IE	DN

Figure 9 - Initial Static Weight Calculations and Trait Distribution

The rescaling of adjusted weights to establish the lowest trait at exactly 1.0 serves multiple methodological purposes:

- **Establishes a clear baseline:** Makes the weighting system intuitive by providing a standard reference point.
- **Preserves mathematical relationships:** Maintains the critical mathematical differences between trait influences.
- **Improves practical implementation:** Facilitates integration into the algorithm and eases performance interpretation.

While this static model effectively addressed trait imbalance, it lacked the ability to adapt to shifting audience dynamics over time. To address this, we introduced a more responsive, evolving system.

PHASE 2.2 DYNAMIC WEIGHTING (FINAL MODEL)

Recognizing the limitations of static correction, we adopted a dynamic weighting strategy. Rather than using fixed weights, the algorithm recalculates weights daily based on the **previous day's dominant trait**:

Previous Dominant Trait	DN Weight	IE Weight	SO Weight
DN	0.85	1.00	1.20
IE	0.52	0.75	1.20
SO	0.52	1.00	1.00

This proportional suppression and compensation mechanism ensures that no single trait repeatedly dominates across performances. The algorithm gradually shifts influence toward underrepresented traits, enhancing diversity in nightly experiences.

These weight combinations were derived through iterative testing on initial data and calibrated to achieve balanced long-term outcomes.

Dates	DN Weight	IE Weight	SO Weight	Weighted AVG - DN	Weighted AVG - IE	Weighted AVG - SO	First	Second	Third
2025-03-05 0:00	1.00	1.00	1.00	6.31	4.58	3.46	DN	IE	SO
2025-03-06 0:00	0.85	1	1.2	5.17	4.62	4.34	DN	IE	SO
2025-03-07 0:00	0.85	1	1.2	5.08	5.08	3.83	IE	DN	SO
2025-03-08 0:00	0.52	0.75	1.2	3.26	3.27	4.17	SO	IE	DN
2025-03-09 0:00	0.52	1	1	3.37	4.20	3.80	IE	SO	DN
2025-03-10 0:00	0.52	0.75	1.2	3.22	3.15	4.96	SO	DN	IE
2025-03-11 0:00	0.52	1	1	3.21	5.11	3.00	IE	DN	SO
2025-03-12 0:00	0.52	0.75	1.2	3.23	3.31	4.51	SO	IE	DN
2025-03-13 0:00	0.52	1	1	3.33	5.11	2.95	IE	DN	SO
2025-03-14 0:00	0.52	0.75	1.2	3.42	3.27	4.53	SO	DN	IE
2025-03-15 0:00	0.52	1	1	2.85	5.48	3.66	IE	SO	DN
2025-03-16 0:00	0.52	0.75	1.2	2.88	3.66	4.80	SO	IE	DN

Figure 10 - Dynamic Weight Calculations and Trait Distribution

Why We Chose Dynamic Over Static:

- Allowed the system to self-correct over time
- Prevented any single trait from dominating across consecutive performances
- Created a more organic, non-repetitive progression of themes

RESULTS

The final dynamic scene generation system was successfully deployed across 12 performances from March 5–16, 2025. Each night’s show featured a distinct scene ordering shaped by that day’s audience input.

The **traits for each performance** were:

March 5th	March 6th	March 7th	March 8th	March 9th	March 10th	March 11th	March 12th	March 13th	March 14th	March 15th	March 16th
DN	DN	IE	SO	IE	SO	IE	SO	IE	SO	IE	SO
IE	IE	DN	IE	SO	DN	DN	IE	DN	DN	SO	IE
SO	SO	SO	DN	DN	IE	SO	DN	SO	IE	DN	DN

Across all 12 performances, the **trait distribution** was:

First position (Dominant Trait):	Second position:	Third position:
DN: 2 times IE: 5 times SO: 5 times	DN: 5 times IE: 5 times SO: 2 times	DN: 5 times IE: 2 times SO: 5 times

Overall Positioning Frequency:

Trait	% as Dominant Trait	% as Second	% as Third
DN	17%	42%	42%
IE	42%	42%	17%
SO	42%	17%	42%

REFLECTION

While the dynamic weighting model succeeded in creating varied and responsive nightly shows, it did not produce the even trait balance originally intended. In fact, Digital Native (DN) was suppressed too heavily — appearing as the dominant trait only 17% of the time — despite being the most naturally prominent trait in the raw audience data.

By comparison, tests with the final data using a static weighting model yielded a more balanced dominance breakdown:

Trait	% as Dominant Trait	% as Second	% as Third
DN	42%	42%	17%
IE	33%	17%	50%
SO	25%	42%	33%

In retrospect, a **calibrated static model** — fine-tuned to produce a ~33% balance — may have achieved more consistent results without sacrificing variation in scene ordering. Although there was initial concern that a static system might lead to repetitive structures, that risk did not materialize during deployment. Instead, the shift to dynamic weighting unintentionally overcompensated for trait imbalance and suppressed DN disproportionately.

TESTING AND DEPLOYMENT

The algorithm was developed and validated over multiple phases:

- **Base Viable Product:** A scene randomizer algorithm was first developed as a minimum viable product in the case that the algorithm was not able to be completed in the timeline.
- **Meeting with Crew:** Meetings were conducted with the writer and marketing team for the play to create questions for the questionnaire as well as information gathering for website design process.
- **Offline Simulation:** A local dataset of audience quiz responses was exported from the SQL database and tested in Excel and PHP to compare weighted average calculations. This ensured alignment between mathematical theory and implementation.
- **Excel Parallel Testing:** Weighted average calculations (static and dynamic) were mirrored in Excel to verify that the logic in the PHP implementation matched expected values. Trait dominance results from the code were compared directly with Excel to validate correctness.
- **Website and Algorithm development:** Web Pages were created for viewer questionnaire and algorithm output.
- **Visual Inspection:** Logs were reviewed regularly during algorithm deployment for anomalies in trait weighting, duplicate scenes, missing "Facts" placement, or uneven depression scene distribution.
- **Product Deployment:** The website and Algorithm were deployed. The first two weeks of Algorithm development were spent testing the Algorithm to ensure that it worked properly with real-time data from users completing the Questionnaire.

PROCESS	JANUARY 06 - FEBRUARY 01				FEBRUARY 02 - MARCH 01				MARCH 02 - MARCH 16			
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 1	Week 2		
Questionnaire Prototype												
Information Gathering												
Hostinger Server Set-up												
Excel Parallel Testing												
Website Development												
Algorithm Development												
Website & Algorithm Testing												
Website Deployment												
Algorithm Deployment												

Figure 11 - Deployment Timeline

TECHNICAL IMPLEMENTATION

The interactive system for Love and Information was built as a fully hosted web-based pipeline deployed on Hostinger web hosting. The platform provided both the PHP execution environment and a built-in MySQL server, making it well-suited for this data-driven theatrical installation.

- **PHP Backend:** A PHP script interfaced with the MySQL database and encapsulated the full logic for trait aggregation, weighting, and scene generation.
- **MySQL Database:** Two tables were used:
 - `quiz_responses`: stored audience responses.
 - `scene_orders`: stored the daily scene ordering and weighted trait values for each date.
- **Scene Order Automation:** A daily job was scheduled to execute the PHP script 4 hours before designated showtimes.

TECHNOLOGIES USED

Technology	Purpose
PHP	<i>Server-side scripting, database querying, dynamic scene generation logic</i>
MySQL	<i>Data storage for survey responses (<code>quiz_responses</code>) and scene orderings (<code>scene_orders</code>)</i>
HTML/CSS	<i>Structure and styling of the webpages, quiz, dashboard, pie chart</i>
JavaScript	<i>Interactivity for survey logic, answer collection, and dynamic user interface updates</i>
SVG	<i>Custom scalable visuals (as seen in Figure 1 – Love and Information Home Page)</i>

THEATRICAL IMPACT OF ALGORITHM

I was invited to attend one of the performances of Love and Information at the Dan School of Music and Drama. Experiencing the show live offered a new lens on the algorithm’s impact beyond the data. The structure of the play—fragmented and fast-moving—reminded me of the digital world we live in today. Each scene felt like a fleeting post on social media.

Experiencing the production live added a completely new dimension to a project that, for much of its development, had been purely computational. When I was building the algorithm, the scenes existed only as lines of text in a database. It was fascinating watching those lines of text change into scenes of the play. Scenes I had randomly sorted hundreds of times in testing suddenly had a “face” to them.

The most compelling part was knowing that the scene ordering was generated specifically for that night’s audience. My own responses—along with those of everyone in the room—shaped the sequence of scenes and the emotional arc we experienced. It was theater not just performed *for* an audience, but partially *by* the audience.

CONCLUSION

This project represented a unique blend of computer science, design, and live performance. Through careful trait modeling, adaptive weighting strategies, and a fully automated web-based deployment system, the algorithm successfully generated a distinct scene order for each of the 12 performances of *Love and Information*.

From a technical standpoint, the project required end-to-end pipeline development involving web design, user data collection, trait scoring, weighted averaging, and dynamic scene ordering. Beyond the code, what made this project truly special was its capacity to shape a shared theatrical experience. Each audience brought a different personality profile into the space, and the algorithm responded by the scene ordering each day.

While dynamic weighting introduced new challenges in achieving balanced trait distribution, it demonstrated the potential of adaptive systems in live art. Had this been a longer run, further refinement could have enhanced trait fairness over time.

Ultimately, the system offered both audience and cast a novel theatrical experience – one in which participation began long before the play started. In a play concerned with humans, emotions, technology and information, using an algorithm to mirror the audience's collective identity was more than a gimmick. It became part of the message.